Text Sentiment Classification Using Deep Learning (RNN/LSTM) 1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task used to determine whether a piece of text expresses positive, negative, or neutral sentiment. It is widely used in applications such as product reviews, social media analysis, and customer feedback systems.

Traditional machine learning models struggle to capture the sequential nature of text. Deep learning models such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks are designed to handle sequential data effectively.

2. Dataset Description

The IMDB Movie Reviews dataset is used for this experiment. It contains 50,000 movie reviews labeled as positive or negative.

- Total samples: 50,000
- Training samples: 25,000
- Testing samples: 25,000
- Balanced dataset (50% positive, 50% negative)

3. Data Preprocessing

3.1 Text Cleaning

- Converted all text to lowercase
- Removed unnecessary symbols and punctuation

3.2 Tokenization

Text is converted into sequences of integers where each word is assigned a unique index.

Example:
"I love this movie" → [12, 45, 67, 89]

3.3 Padding

Sequences are padded to ensure equal length.

Example:
[1, 2, 3] → [0, 0, 1, 2, 3]

4. Model Architectures

4.1 RNN Model

- Embedding layer (128 dimensions)
- SimpleRNN layer (64 units)
- Dense output layer with sigmoid activation

4.2 LSTM Model

- Embedding layer (128 dimensions)
- LSTM layer (64 units)
- Dense output layer with sigmoid activation

5. Experimental Setup

- Vocabulary size: 10,000
- Sequence length: 200
- Batch size: 64
- Epochs: 5
- Optimizer: Adam
- Loss function: Binary Crossentropy

6. Results and Analysis

6.1 Performance Metrics

- Accuracy
- Precision
- Recall
- F1-score

6.2 Observations

- LSTM achieved higher accuracy than RNN
- RNN struggled with long-term dependencies
- Increasing sequence length improved performance slightly
- Larger embedding size improved representation

7. Graph Analysis

- Training and validation loss curves show LSTM is more stable
- Accuracy curves show better convergence for LSTM

8. RNN vs LSTM Feature RNN LSTM

Memory Short-term Long-term

Gradient Issue Yes Reduced

Accuracy Lower Higher

LSTM uses gates (input, forget, output) which help it retain important information over long sequences.

9. Conclusion

In this experiment, both RNN and LSTM models were implemented for sentiment classification. The LSTM model outperformed the RNN model in terms of accuracy and stability. This demonstrates the effectiveness of LSTM in capturing long-term dependencies in text data.

10. Future Work

- Implement GRU for comparison
- Use pre-trained embeddings (Word2Vec, GloVe)
- Apply model on real-world datasets (Twitter, Amazon reviews)